Finominal Quant Challenge

Junior Quantitative Analyst

Momentum Factor Optimization

Field	Details
Role	**Junior Quantitative Analyst**
Time Limit	**3 hrs**
Difficulty	**Intermediate**

What You Will Receive

Along with this document, you will be provided:

- Historical daily closing price data for 30 stocks.
- Historical market capitalization data for the same 30 stocks.

1. Objective

Optimize a monthly-rebalanced, market-capitalization-weighted momentum strategy to maximize the Sharpe ratio by identifying the optimal lookback period.

2. Timeline and Execution Rules

- Review the data.
- **Backtest Period**: The backtest execution, portfolio construction, and performance tracking strictly begin on the **first trading day of 2006**. Price data prior to 2006 is used exclusively for calculating the initial momentum scores.
- The backtest must extend through the final available trading day, with all performance metrics calculated relative to this terminal date.
- **Rebalance Date (T_{reb})**: The **first trading day** of every month. All trades to adjust the portfolio to target weights are executed at the market close on this date.
- **Calculation Date (T_{calc})**: The trading day immediately preceding the Rebalance Date (typically the last trading day of the prior month).
- *Constraint*: All momentum scores and market capitalization weights **must** be calculated using the closing price and market capitalization data exactly on T_{calc}.

3. Signal Generation (Momentum)

On each T_{calc}, calculate the momentum score M_i for all available stocks over a lookback period expressed in exact trading days (L_{days}):

$$M_i = \frac{P_{i,T_{calc}}}{P_{i,T_{calc}-L_{days}}} - 1$$

- **Ranking**: Rank all stocks in descending order based on their calculated M_i (Rank 1 = highest momentum score).

4. Portfolio Construction Rules

A. Selection and Buffer Constraint

The portfolio must hold exactly **10 stocks** at all times. On each Rebalance Date, apply the following logic in order:

1. **Evaluate**: Check the new momentum rank of the stocks currently held in the portfolio.
2. **Exit**: Remove any existing portfolio stock if its new rank is > 12.
3. **Retain**: Keep any existing portfolio stock if its new rank is ≤ 12.
4. **Addition**: Fill all vacant slots (to reach exactly 10 stocks) with the highest-ranked stocks not currently held in the portfolio.

B. Weighting and 20% Capping Algorithm

1. **Initial Weight**: Calculate the raw market-cap weight for the 10 selected stocks using data strictly from T_{calc}:

$$w_{i,raw} = \frac{MC_{i,T_{calc}}}{\sum_{j=1}^{10} MC_{j,T_{calc}}}$$

2. **Capping & Scaling**: Iteratively enforce a **20%** maximum weight constraint:
   - Cap any stock where $w_i > 0.20$ at exactly **0.20**.
   - Calculate the total excess weight: $E = \sum \left(w_{original} - 0.20\right)$ collected from all capped stocks.
   - Distribute E among the remaining uncapped stocks proportionally, based on their relative market capitalizations.
   - Repeat this capping and redistribution process until no single stock's weight exceeds **0.20** and the sum of all weights equals **1.00** (**100%**).

5. Optimization and Performance Metrics

- **Optimization Goal**: Maximize the Annualized Sharpe Ratio.

- **Lookback Parameter Space**: Test the strategy independently across the following exact trading day lookbacks (L_{days}):
    - **21, 63, 126, 189, and 252 days**
- **Metrics (Calculated using Daily Returns)**:
    - **Annualized Sharpe Ratio**:

$$SR = \frac{\mu_{p,daily}}{\sigma_{p,daily}} \times \sqrt{252}$$

*(where $\mu_{p,daily}$ is the mean of the daily portfolio returns, and $\sigma_{p,daily}$ is the sample standard deviation of the daily portfolio returns, assuming a **0%** risk-free rate).*

    - **Cumulative Return**:

$$CR = \left(\prod_{t=1}^{N} \left(1 + R_{p,t,daily}\right) \right) - 1$$

(where $R_{p,t,daily}$ is the daily portfolio return on day t, and N is the total number of trading days in the backtest starting from the first trading day of 2006).

6. Output Report Format

Deliverable	Details
Python Code	Clean, readable, and well-commented code. A README.md file explaining project setup and execution.
Results	Share the following four performance metrics: • Optimal Lookback Period (Days) • Maximum Annualized Sharpe Ratio • Cumulative Return (%) • CAGR (%)
Loom Video	A short screen recording (3-5 mins) walking through your approach, code, and a qualitative, critical assessment of the trading strategy.

Your Loom video is your chance to explain your thinking. We care about how you approach problems, not just the final output. Narrate your decisions, trade-offs, and any shortcuts you took due to time constraints.